

Construction Progress & Customer Experience Platform

Project Proposal & Scope Document

1. Executive Summary

This proposal outlines the development of a modern cloud-based Construction Progress & Customer Experience Platform designed specifically for Australian residential home builders.

The platform aims to improve communication, transparency, customer satisfaction, operational efficiency, and post-handover warranty management throughout the home construction lifecycle.

Unlike traditional construction management systems that primarily focus on internal project operations, this platform is designed around the customer journey while also providing supervisors, trades, and builders with practical tools to manage projects more efficiently.

The system will provide:

- Real-time construction progress tracking
- Timeline-based activity updates
- Photo and document management
- Customer notifications
- Invoice and billing notifications
- Warranty and defect tracking
- FAQ and AI-powered customer assistance
- Multi-tenant builder management
- Auto-scalable cloud infrastructure

The platform will be designed as a SaaS (Software as a Service) solution targeting volume home builders, project builders, and residential construction companies.

2. Business Objectives

Primary Objectives

- Improve customer experience during home construction
- Reduce customer support calls and email traffic
- Increase transparency between builders and customers
- Centralize project communication and updates
- Streamline post-handover defect and warranty management
- Provide a scalable platform for residential builders

Secondary Objectives

- Create recurring SaaS revenue
 - Enable enterprise builder onboarding
 - Support future integrations with accounting and ERP systems
 - Enable AI-assisted customer support
 - Build a scalable multi-tenant architecture
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3. Problem Statement

Residential builders and customers commonly experience:

- Limited visibility into project progress
- Repetitive customer calls and emails
- Poor update frequency
- Unclear construction timelines
- Disconnected communication channels
- Difficulty managing warranty issues post-handover
- Lack of centralized document access
- Delays caused by inefficient communication

Most existing construction management systems are focused on internal operations and are overly complex for residential customer engagement.

This platform aims to solve these problems through a simple, visual, customer-focused experience.

4. Proposed Solution

The proposed solution is a cloud-native Construction Progress & Customer Experience Platform consisting of:

Customer Portal

Customers can:

- View live construction progress
- Track stage completion
- Access timeline updates
- View site photos and documents
- Receive notifications
- Access invoices and payment reminders
- Log warranty and maintenance issues
- Access FAQ and AI assistant

Builder Management Portal

Builders and supervisors can:

- Manage multiple projects
- Update project stages
- Upload site photos and updates
- Communicate with customers
- Assign and track warranty issues
- Configure notifications
- Manage FAQs and AI responses
- View operational dashboards

Trade Access Portal

Trades can:

- View assigned jobs
- Upload completion evidence
- Update job status
- Report issues or delays

5. Key Features

5.1 Project Progress Tracking

- Construction stage tracking
- Percentage-based progress indicators
- Timeline and milestone tracking
- Estimated completion forecasting
- Delay indicators

Example Stages

- Site Cut
 - Slab
 - Frame
 - Lockup
 - Fixing
 - Completion
 - Handover
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5.2 Activity Timeline

Visual timeline feed displaying:

- Supervisor updates
- Construction milestones
- Photos
- Inspection updates
- Delays
- Site activities

Features:

- Chronological activity feed
 - Photo attachments
 - Tagged updates
 - Customer comments
 - Supervisor replies
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5.3 Notifications System

Notification types include:

Progress Notifications

- Stage started/completed
- Inspection scheduled
- New site photos uploaded
- Delays recorded

Financial Notifications

- Invoice issued
- Payment due reminders
- Payment confirmations

Support Notifications

- Question answered
- Warranty issue updated
- AI assistant responses

Delivery methods:

- Push notifications
- Email notifications
- In-app notifications

5.4 Warranty & Defect Management

Post-handover functionality includes:

- Warranty issue submission
- Photo-based defect reporting
- Trade assignment
- Status tracking
- Scheduled maintenance reminders
- Warranty expiry reminders

Workflow:

Reported → Assigned → Scheduled → In Progress → Resolved

5.5 FAQ & AI Assistant

FAQ System

Builder-managed knowledge base covering:

- Construction stages
- Delays
- PCI process
- Site access
- Payments
- Variations
- Warranty information

AI Assistant

AI-powered assistant capable of answering:

- General construction questions
- Builder policy questions
- Project-specific questions
- Timeline and stage explanations

The AI assistant will use:

- FAQ content
- Project data
- Uploaded documents
- Builder-defined policies

AI guardrails will ensure responses are limited to approved information sources.

5.6 Document Management

Supported documents include:

- Floor plans
- Contracts
- Invoices
- Certificates
- Appliance manuals
- Warranty documents

- PCI checklists

Features:

- Secure uploads
 - Role-based access
 - Version tracking
 - Cloud storage
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6. User Roles & Use Cases

The platform will support the following primary user roles. Access will be controlled using role-based permissions within each tenant/company.

6.1 Customer

Customers are home buyers or property owners who use the platform to track the construction journey and communicate with the builder.

Key Use Cases

- View current construction stage and progress percentage
 - View project timeline and latest site activity
 - Receive notifications for milestones, invoices, inspections, delays, and handover updates
 - View uploaded site photos and documents
 - Ask project-related questions through the portal or AI assistant
 - Access FAQs related to construction stages, payments, PCI, handover, and warranty
 - Submit warranty or maintenance issues after handover
 - Upload photos for reported defects
 - Track defect or warranty issue status
 - Access handover documents, appliance manuals, certificates, and warranty information
 - Receive reminders for warranty periods and follow-up inspections
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6.2 Site Supervisor

Site supervisors manage day-to-day project updates and are responsible for keeping customers informed.

Key Use Cases

- View assigned projects and site status
 - Update project stages such as slab, frame, lockup, fixing, completion, and handover
 - Upload site photos and progress notes
 - Create timeline updates for customers
 - Log delays due to weather, materials, inspections, or trade availability
 - Respond to customer questions
 - Upload key documents or inspection updates
 - Assign warranty or defect issues to relevant trades
 - Track defect resolution progress
 - Receive reminders for pending customer questions or overdue updates
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6.3 Trades

Trades are contractors or subcontractors such as electricians, plumbers, carpenters, painters, tilers, and landscapers.

Key Use Cases

- View assigned jobs or warranty tasks
 - Access relevant site details and job instructions
 - Update task status such as scheduled, started, completed, or blocked
 - Upload completion photos or evidence of work
 - Report issues such as missing materials, access problems, incorrect drawings, or unsafe conditions
 - Comment on assigned tasks
 - Receive notifications for upcoming work, changes, or urgent issues
 - Support warranty issue resolution after handover
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6.4 Company Admin

Company admins manage system configuration and operational setup for each builder/company tenant.

Key Use Cases

- Manage company profile and branding
 - Configure company logo, colours, email templates, and customer-facing portal branding
 - Manage users, roles, and permissions
 - Invite customers, supervisors, trades, and management users
 - Create and manage projects/sites
 - Assign supervisors to projects
 - Configure construction stages and milestone templates
 - Manage FAQ content and approved AI knowledge base
 - Configure notification templates and notification rules
 - Upload standard documents and policy templates
 - Manage billing settings and subscription/site licensing
 - Manage translation/localisation settings
 - View audit logs and user activity
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6.5 Company Management

Company management users require higher-level visibility across projects, teams, customer experience, and business performance.

Key Use Cases

- View executive dashboard across all active projects
 - Monitor project counts by stage, region, supervisor, and status
 - Identify delayed or at-risk projects
 - View customer communication trends and unanswered questions
 - Track open warranty and defect issues
 - View supervisor workload and update frequency
 - Review operational reports and analytics
 - Monitor billing and site usage
 - Access audit and compliance reporting
 - Review AI/FAQ usage trends and common customer questions
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7. Technical Architecture

Frontend

Web Portal

- Next.js
- TypeScript
- Tailwind CSS

Mobile Application

- React Native
 - Expo
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Backend

Core Technologies

- Java 21
- Spring Boot 3
- Spring Security
- Spring Data JPA
- Spring Batch
- PostgreSQL

Architecture Style

- Modular monolith
 - Containerized deployment
 - Stateless API services
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Cloud Infrastructure

AWS Services

- ECS Fargate
- Application Load Balancer
- Aurora PostgreSQL
- S3
- EventBridge
- SQS
- CloudFront
- Cognito
- CloudWatch

- WAF
- KMS

Scalability

- Auto-scaling container infrastructure
 - Multi-AZ deployment
 - Stateless service architecture
 - Asynchronous processing via queues
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Authentication & Security

- Amazon Cognito
 - Microsoft Entra ID federation
 - JWT-based authentication
 - Role-based access control
 - Multi-tenant isolation
 - Encrypted storage
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8. Billing Platform

The system will include a billing platform supporting:

- Per-site billing
- Subscription billing
- Volume pricing
- Invoice generation
- Renewal processing
- Payment tracking
- Reporting

Planned Billing Features

- Invoice management
- Automated renewal reminders
- Batch invoice generation
- Discount management
- GST handling

- Financial reporting

Technologies

- Spring Batch
 - Stripe integration
 - Scheduled batch processing
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9. Multi-Tenant SaaS Design

The platform will support multiple builder organizations using a shared SaaS platform architecture.

Each organization will operate as a separate tenant with isolated:

- Projects
- Users
- Documents
- Notifications
- FAQ content
- Warranty records
- Billing data
- Branding configuration

The architecture will support future enterprise scaling.

Multi-Tenant Requirements

- Every company is represented as a tenant/organization
- All key data tables include `tenant_id` or `organization_id`
- Backend APIs enforce tenant-level isolation
- Users can only access data within their assigned organization
- Platform administrators can manage all tenants
- Audit logs record tenant, user, action, and timestamps

White-Label Branding

Each tenant/company can configure:

- Company logo

- Brand colours
- Customer portal theme
- Branded emails
- Branded PDFs and reports
- Company-specific support details
- Optional custom subdomain support

Locale & Translation Support

The platform will support localisation and multilingual capabilities.

Locale Features

- Default locale per company
- Locale preference per user
- Time zone support
- Locale-aware date, time, and currency formatting
- Australian English as default initial locale

Translation Features

- Translation key-based UI design
 - Multi-language support for UI labels
 - Translated notification templates
 - Translated email templates
 - Translated FAQ content
 - Fallback language support
 - Future support for additional languages
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10. Project Scope

Included in Scope

Phase 1 – MVP

- User authentication
- Organization management
- Project management
- Stage progress tracking

- Activity timeline
- Photo uploads
- Customer portal
- Notifications
- Basic admin dashboard

Phase 2

- Warranty management
- Document management
- Trade portal
- Enhanced reporting
- Mobile optimization

Phase 3

- FAQ system
 - AI assistant
 - Advanced analytics
 - Billing platform enhancements
 - Enterprise features
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Out of Scope (Initial Release)

- Full accounting platform
 - Payroll processing
 - Complex scheduling engine
 - ERP replacement
 - Procurement management
 - BIM integration
 - Advanced AI automation
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11. Delivery Approach

Methodology

- Agile delivery
- Two-week sprints

- Incremental releases
- Continuous feedback cycles

Team Structure

- Project Manager / Technical Lead
 - Business Analyst
 - Backend Developers
 - Frontend Developers
 - QA Engineer
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12. Proposed Timeline

Phase	Duration	Deliverables
Discovery & Architecture	1 Month	Requirements, architecture, UI planning
MVP Development	3–4 Months	Core platform, progress tracking
Builder Portal Expansion	2–3 Months	Dashboards, notifications, documents
Warranty Module	2 Months	Defect management
AI & FAQ Features	2 Months	AI assistant, FAQ platform
Stabilization & Pilot Rollout	1 Month	Testing, onboarding
Estimated Total Timeline: 10–12 Months		

13. Expected Business Benefits

For Builders

- Reduced customer support workload
- Improved operational visibility
- Better customer satisfaction
- Centralized communication
- Streamlined defect tracking
- Improved professional image

For Customers

- Better transparency
 - Reduced uncertainty
 - Real-time updates
 - Easier communication
 - Centralized access to project information
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14. Commercial Model

Suggested pricing models:

Per Site Pricing

- Standard: \$99 per site (12 months)
- Volume discounts available

Subscription Pricing

- Monthly builder subscriptions
- Enterprise pricing available

Optional Add-ons

- AI assistant
 - White-label branding
 - Advanced analytics
 - Extended warranty management
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15. Future Enhancements

Potential future features include:

- AI-powered project summaries
 - Predictive delay analysis
 - Builder benchmarking
 - Customer satisfaction analytics
 - Accounting integrations
 - Voice/photo AI analysis
 - Smart scheduling
 - Trade performance tracking
 - Mobile offline support
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16. Conclusion

The proposed Construction Progress & Customer Experience Platform addresses significant gaps in residential construction communication and customer engagement.

By combining project visibility, timeline-based updates, notifications, warranty management, and AI-assisted support within a scalable SaaS platform, the solution has strong commercial potential within the Australian residential construction industry.

The platform is designed to support both operational efficiency for builders and a significantly improved experience for customers throughout the home construction lifecycle.